

ECS 409/609

Assignment 7

Conditional Construct

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For code repository of this assignment:  MIPS Assignments

1. Problem 1

Write a MIPS assembly program that takes two number (can be anything oating point or integer) as an input and print maximum between two of them as follows:

32.6 is greater than 25.0

MIPS Assembly Source

```
1 # Assignment 7: Conditional Construct
2 # Programmed By : Kunwar Arpit Singh
3
4 .data
5 # --- Strings for Prompts and Output ---
6 prompt1: .asciiz "Programmed By: Kunwar Arpit Singh 22185\n\nEnter the first
7             floating-point number: "
8 prompt2: .asciiz "Enter the second floating-point number: "
9 is_greater: .asciiz "is greater than "
10 newline: .asciiz "\n"
11
12 .text
13 .globl main
14 main:
15     li $v0, 4
16     la $a0, prompt1
17     syscall
18
19     li $v0, 6
20     syscall
21
22     mov.s $f1, $f0
23
24     li $v0, 4
25     la $a0, prompt2
26     syscall
27
```

```

28     li $v0, 6
29     syscall
30
31     c.lt.s $f0, $f1
32
33     bc1t num1_is_greater
34
35     li $v0, 2
36     mov.s $f12, $f0
37     syscall
38
39     li $v0, 4
40     la $a0, is_greater
41     syscall
42
43     li $v0, 2
44     mov.s $f12, $f1
45     syscall
46
47     j exit
48
49 num1_is_greater:
50
51     li $v0, 2
52     mov.s $f12, $f1
53     syscall
54
55     li $v0, 4
56     la $a0, is_greater
57     syscall
58
59     li $v0, 2
60     mov.s $f12, $f0
61     syscall
62
63 exit:
64     li $v0, 4
65     la $a0, newline
66     syscall
67
68     li $v0, 10
69     syscall

```

Output Screenshots

Integer Register values and Console

The screenshot shows the 'Int Regs [10]' window and the 'Console' window. The 'Int Regs [10]' window displays the following values:

Register	Value
PC	4194500
EPC	0
Cause	0
BadVAddr	0
Status	805371664
HI	0
LO	0
R0 [r0]	0
R1 [at]	268500992
R2 [v0]	10
R3 [v1]	0
R4 [a0]	268501132
R5 [a1]	2147479220
R6 [a2]	2147479240
R7 [a3]	0
R8 [t0]	0
R9 [t1]	0
R10 [t2]	0
R11 [t3]	0
R12 [t4]	0
R13 [t5]	0
R14 [t6]	0
R15 [t7]	0
R16 [s0]	0
R17 [s1]	0
R18 [s2]	0
R19 [s3]	0
R20 [s4]	0
R21 [s5]	0
R22 [s6]	0
R23 [s7]	0
R24 [t8]	0
R25 [t9]	0
R26 [k0]	0
R27 [k1]	0
R28 [gp]	268468224
R29 [sp]	2147479216
R30 [s8]	0
R31 [ra]	4194328

The 'Console' window displays the following output:

```
Programmed By: Kunwar Arpit Singh 22185
Enter the first floating-point number: 36.20
Enter the second floating-point number: 67.98
67.98000336 is greater than 36.20000076
```

Floating Point Register values and Console

The screenshot shows the 'FP Regs' window and the 'Console' window. The 'FP Regs' window displays the following values:

Register	Value
FIR	38912
PCSR	0
Single Precision	
FG0	0.000000
FG1	0.000000
FG2	0.000000
FG3	0.000000
FG4	0.000000
FG5	0.000000
FG6	0.000000
FG7	0.000000
FG8	0.000000
FG9	0.000000
FG10	0.000000
FG11	0.000000
FG12	0.000000
FG13	0.000000
FG14	0.000000
FG15	0.000000
FG16	0.000000
FG17	0.000000
FG18	0.000000
FG19	0.000000
FG20	0.000000
FG21	0.000000
FG22	0.000000
FG23	0.000000
FG24	0.000000
FG25	0.000000
FG26	0.000000
FG27	0.000000
FG28	0.000000
FG29	0.000000
FG30	0.000000
FG31	0.000000
Double Precision	
FD0	0.000000
FD2	0.000000
FD4	0.000000
FD6	0.000000
FD8	0.000000
FD10	0.000000
FD12	0.000000
FD14	0.000000
FD16	0.000000
FD18	0.000000
FD20	0.000000
FD22	0.000000
FD24	0.000000
FD26	0.000000
FD28	0.000000
FD30	0.000000

The 'Console' window displays the following output:

```
Programmed By: Kunwar Arpit Singh 22185
Enter a year: 2020
Input year is a leap year
```

2. Problem 2

Write an assembly program that takes year as an input from the user and check whether the input year is leap year or not. If it is leap year prompt the message

Input year is a leap year

Otherwise, prompt the message

Input year is not a leap year

MIPS Assembly Source

```
1 # Assignment 7: Conditional Construct
2 # Programmed By : Kunwar Arpit Singh
3
4 .data
5     prompt_year: .asciiz "Programmed By: Kunwar Arpit Singh 22185\n\nEnter a
        year:"
6     message_is_leap: .asciiz "Input year is a leap year\n"
7     message_not_leap: .asciiz "Input year is not a leap year\n"
8
9 .text
10 .globl main
11
12 main:
13     li $v0, 4
14     la $a0, prompt_year
15     syscall
16
17     li $v0, 5
18     syscall
19
20     move $s0, $v0
21
22     li $t1, 4
23     div $s0, $t1
24     mfhi $t2
25
26     bne $t2, $zero, not_leap
27
28     li $t1, 100
29     div $s0, $t1
30     mfhi $t2
31
32     bne $t2, $zero, is_leap
33
34     li $t1, 400
35     div $s0, $t1
36     mfhi $t2
37
```

```
38     bne $t2, $zero, not_leap
39
40 is_leap:
41     li $v0, 4
42     la $a0, message_is_leap
43     syscall
44     j exit
45
46 not_leap:
47     li $v0, 4
48     la $a0, message_not_leap
49     syscall
50
51 exit:
52     li $v0, 10
53     syscall
```

Output Screenshots

Integer Register values and Console

The image displays two screenshots of a debugger interface, showing the state of integer registers and the console output for a program that checks if a year is a leap year.

Top Screenshot:

- Integer Registers (Int Regs [10]):**
 - PC = 4194452
 - EPC = 0
 - Cause = 0
 - BadVAddr = 0
 - Status = 805371664
 - HI = 20
 - LO = 20
 - R0 [r0] = 0
 - R1 [at] = 268500992
 - R2 [v0] = 10
 - R3 [v1] = 0
 - R4 [a0] = 268501048
 - R5 [a1] = 2147479220
 - R6 [a2] = 2147479240
 - R7 [a3] = 0
 - R8 [t0] = 0
 - R9 [t1] = 100
 - R10 [t2] = 20
 - R11 [t3] = 0
 - R12 [t4] = 0
 - R13 [t5] = 0
 - R14 [t6] = 0
 - R15 [t7] = 0
 - R16 [s0] = 2020
 - R17 [s1] = 0
 - R18 [s2] = 0
 - R19 [s3] = 0
 - R20 [s4] = 0
 - R21 [s5] = 0
 - R22 [s6] = 0
 - R23 [s7] = 0
 - R24 [t8] = 0
 - R25 [t9] = 0
 - R26 [k0] = 0
 - R27 [k1] = 0
 - R28 [gp] = 268468224
 - R29 [sp] = 2147479216
 - R30 [s8] = 0
 - R31 [ra] = 4194328
- Console:**
 - Programmed By: Kunwar Arpit Singh 22185
 - Enter a year: 2020
 - Input year is a leap year
- Text:**
 - Kernel Text Segment [80000000]..[80010000]
 - [80000180] 0001d821 addu \$27, \$0, \$1 ; 90: move \$k1 \$at # Save \$at

Bottom Screenshot:

- Integer Registers (Int Regs [10]):**
 - PC = 4194452
 - EPC = 0
 - Cause = 0
 - BadVAddr = 0
 - Status = 805371664
 - HI = 3
 - LO = 505
 - R0 [r0] = 0
 - R1 [at] = 268500992
 - R2 [v0] = 10
 - R3 [v1] = 0
 - R4 [a0] = 268501075
 - R5 [a1] = 2147479220
 - R6 [a2] = 2147479240
 - R7 [a3] = 0
 - R8 [t0] = 0
 - R9 [t1] = 4
 - R10 [t2] = 3
 - R11 [t3] = 0
 - R12 [t4] = 0
 - R13 [t5] = 0
 - R14 [t6] = 0
 - R15 [t7] = 0
 - R16 [s0] = 2023
 - R17 [s1] = 0
 - R18 [s2] = 0
 - R19 [s3] = 0
 - R20 [s4] = 0
 - R21 [s5] = 0
 - R22 [s6] = 0
 - R23 [s7] = 0
 - R24 [t8] = 0
 - R25 [t9] = 0
 - R26 [k0] = 0
 - R27 [k1] = 0
 - R28 [gp] = 268468224
 - R29 [sp] = 2147479216
 - R30 [s8] = 0
 - R31 [ra] = 4194328
- Console:**
 - Programmed By: Kunwar Arpit Singh 22185
 - Enter a year: 2023
 - Input year is not a leap year
- Text:**
 - Kernel Text Segment [80000000]..[80010000]
 - [80000180] 0001d821 addu \$27, \$0, \$1 ; 90: move \$k1 \$at # Save \$at

Floating Point Register values and Console

The image displays a MIPS simulator interface with two screenshots. The top screenshot shows the initial state of the floating point registers (FIR = 38912, FCSR = 0) and a console window with the following text:

```
Programmed By: Kunwar Arpit Singh 22185
Enter the first floating-point number: 36.20
Enter the second floating-point number: 67.98
67.98000336 is greater than 36.20000076
```

The bottom screenshot shows the same simulator with the console window displaying:

```
Programmed By: Kunwar Arpit Singh 22185
Enter a year: 2023
Input year is not a leap year
```

The floating point register values are listed in two sections: Single Precision and Double Precision. The Single Precision registers (FG0 to FG31) are all 0.000000, except for FG12 which is 36.200001. The Double Precision registers (FP0 to FP30) are all 0.000000, except for FP0 which is 18038870177.990002.

The console window also displays the following text:

```
Kernel Text Segment [80000000]..[80010000]
[80000180] 0001d821 addu $27, $0, $1 ; 90: move $k1 $at # Save $at
[80000184] 3c019000 lui $1, -28672 ; 92: sw $v0 $1 # Not re-entrant and we can't trust $sp
[80000188] ac220200 sw $2, 512($1)
[8000018c] 3c019000 lui $1, -28672 ; 93: sw $a0 $2 # But we need to use these registers
```

3. Problem 3

Problem statement

Write an assembly program that determines whether the student is allowed to sit the examination provided his/her attendance is 75%. For the given problem statement, the MIPS assembly program takes the following input: The name of student, Total number of class held and Total class attended by the student. The output format is as follows:

Ajay is allowed to sit in the exam.
or
Ajay is not allowed to sit in the exam.

MIPS Assembly Source

```
1 # Assignment 7: Conditional Construct
2 # Programmed By : Kunwar Arpit Singh
3
4 .data
5     prompt_name: .asciiz "Programmed_By:_Kunwar_Arpit_Singh_22185\n\nEnter_
6         student_name:_"
7     prompt_total: .asciiz "Enter_total_number_of_classes_held:_"
8     prompt_attended: .asciiz "Enter_number_of_classes_attended:_"
9     is_allowed: .asciiz "_is_allowed_to_sit_in_the_exam.\n"
10    not_allowed: .asciiz "_is_not_allowed_to_sit_in_the_exam.\n"
11
12    student_name: .space 256
13
14 .text
15 .globl main
16 main:
17     li $v0, 4
18     la $a0, prompt_name
19     syscall
20
21     li $v0, 8
22     la $a0, student_name
23     li $a1, 256
24     syscall
25
26     la $t0, student_name
27
28 find_newline_loop:
29     lb $t1, 0($t0)
30     beq $t1, 10, replace_newline
31     beq $t1, 0, done_checking
32     addi $t0, $t0, 1
33     j find_newline_loop
34
35 replace_newline:
```



```

36     sb $zero, 0($t0)
37
38 done_checking:
39     li $v0, 4
40     la $a0, prompt_total
41     syscall
42
43     li $v0, 5
44     syscall
45     move $s0, $v0
46
47     li $v0, 4
48     la $a0, prompt_attended
49     syscall
50
51     li $v0, 5
52     syscall
53     move $s1, $v0
54
55     li $t0, 100
56     mult $s1, $t0
57     mflo $t2
58
59     li $t0, 75
60     mult $s0, $t0
61     mflo $t3
62
63     blt $t2, $t3, not_allowed_label
64
65     li $v0, 4
66     la $a0, student_name
67     syscall
68
69     li $v0, 4
70     la $a0, is_allowed
71     syscall
72
73     j exit
74
75 not_allowed_label:
76     li $v0, 4
77     la $a0, student_name
78     syscall
79
80     li $v0, 4
81     la $a0, not_allowed
82     syscall
83
84 exit:
85     li $v0, 10
86     syscall

```

Output Screenshots

Integer Register values and Console

The image displays two screenshots of a MIPS simulator interface, showing the state of integer registers and the console output for two different test cases.

Top Screenshot:

- Integer Registers (Int Regs [10]):**
 - PC = 4194568
 - EPC = 0
 - Cause = 0
 - BadVAddr = 0
 - Status = 805371664
 - HI = 0
 - LO = 3000
 - R0 [r0] = 0
 - R1 [at] = 268500992
 - R2 [v0] = 10
 - R3 [v1] = 0
 - R4 [a0] = 268501126
 - R5 [a1] = 256
 - R6 [a2] = 2147479240
 - R7 [a3] = 0
 - R8 [t0] = 75
 - R9 [t1] = 10
 - R10 [t2] = 3100
 - R11 [t3] = 3000
 - R12 [t4] = 0
 - R13 [t5] = 0
 - R14 [t6] = 0
 - R15 [t7] = 0
 - R16 [s0] = 40
 - R17 [s1] = 31
 - R18 [s2] = 0
 - R19 [s3] = 0
 - R20 [s4] = 0
 - R21 [s5] = 0
 - R22 [s6] = 0
 - R23 [s7] = 0
 - R24 [t8] = 0
 - R25 [t9] = 0
 - R26 [k0] = 0
 - R27 [k1] = 0
 - R28 [gp] = 268468224
 - R29 [sp] = 2147479216
 - R30 [s8] = 0
 - R31 [ra] = 4194328
- Console:**

```
Programmed By: Kunwar Arpit Singh 22185
Enter student name: Arpit
Enter total number of classes held: 40
Enter number of classes attended: 31
Arpit is allowed to sit in the exam.
```
- Assembly Code (Visible at the bottom):**

```
[00400080] 00028021 addu $t6, $0, $2 ; 45: move $s0, $v0
[00400084] 34020004 ori $2, $0, 4 ; 47: li $v0, 4
[00400088] 3c011001 lui $1, 4097 [prompt_attended]; 48: la $a0, prompt_attended
[0040008c] 34240063 ori $4, $1, 99 [prompt_attended]
[00400090] 0000000c syscall ; 49: syscall
[00400094] 34020005 ori $2, $0, 5 ; 51: li $v0, 5
[00400098] 0000000c syscall ; 52: syscall
[0040009c] 00028821 addu $t7, $0, $2 ; 53: move $s1, $v0
[004000a0] 34080064 ori $8, $0, 100 ; 55: li $t0, 100
```

Bottom Screenshot:

- Integer Registers (Int Regs [10]):**
 - PC = 4194568
 - EPC = 0
 - Cause = 0
 - BadVAddr = 0
 - Status = 805371664
 - HI = 0
 - LO = 3000
 - R0 [r0] = 0
 - R1 [at] = 268500992
 - R2 [v0] = 10
 - R3 [v1] = 0
 - R4 [a0] = 268501159
 - R5 [a1] = 256
 - R6 [a2] = 2147479240
 - R7 [a3] = 0
 - R8 [t0] = 75
 - R9 [t1] = 10
 - R10 [t2] = 2800
 - R11 [t3] = 3000
 - R12 [t4] = 0
 - R13 [t5] = 0
 - R14 [t6] = 0
 - R15 [t7] = 0
 - R16 [s0] = 40
 - R17 [s1] = 28
 - R18 [s2] = 0
 - R19 [s3] = 0
 - R20 [s4] = 0
 - R21 [s5] = 0
 - R22 [s6] = 0
 - R23 [s7] = 0
 - R24 [t8] = 0
 - R25 [t9] = 0
 - R26 [k0] = 0
 - R27 [k1] = 0
 - R28 [gp] = 268468224
 - R29 [sp] = 2147479216
 - R30 [s8] = 0
 - R31 [ra] = 4194328
- Console:**

```
Programmed By: Kunwar Arpit Singh 22185
Enter student name: Ajay
Enter total number of classes held: 40
Enter number of classes attended: 28
Ajay is not allowed to sit in the exam.
```
- Assembly Code (Visible at the bottom):**

```
[004000e8] 3c011001 lui $1, 4097 [student_name]; 77: la $a0, student_name
[004000ec] 342400cc ori $4, $1, 204 [student_name]
[004000f0] 0000000c syscall ; 78: syscall
[004000f4] 34020004 ori $2, $0, 4 ; 80: li $v0, 4
[004000f8] 3c011001 lui $1, 4097 [not_allowed]; 81: la $a0, not_allowed
[004000fc] 342400a7 ori $4, $1, 167 [not_allowed]
[00400100] 0000000c syscall ; 82: syscall
[00400104] 3402000a ori $2, $0, 10 ; 85: li $v0, 10
[00400108] 0000000c syscall ; 86: syscall
```

Floating Point Register values and Console

FP Regs | **Int Regs [10]** | **Data** | **Text**

FP Regs

FIR = 38912
FCSR = 0

Single Precision

FG0 = 0.000000
FG1 = 0.000000
FG2 = 0.000000
FG3 = 0.000000
FG4 = 0.000000
FG5 = 0.000000
FG6 = 0.000000
FG7 = 0.000000
FG8 = 0.000000
FG9 = 0.000000
FG10 = 0.000000
FG11 = 0.000000
FG12 = 0.000000
FG13 = 0.000000
FG14 = 0.000000
FG15 = 0.000000
FG16 = 0.000000
FG17 = 0.000000
FG18 = 0.000000
FG19 = 0.000000
FG20 = 0.000000
FG21 = 0.000000
FG22 = 0.000000
FG23 = 0.000000
FG24 = 0.000000
FG25 = 0.000000
FG26 = 0.000000
FG27 = 0.000000
FG28 = 0.000000
FG29 = 0.000000
FG30 = 0.000000
FG31 = 0.000000

Double Precision

FP0 = 0.000000
FP2 = 0.000000
FP4 = 0.000000
FP6 = 0.000000
FP8 = 0.000000
FP10 = 0.000000
FP12 = 0.000000
FP14 = 0.000000
FP16 = 0.000000
FP18 = 0.000000
FP20 = 0.000000
FP22 = 0.000000
FP24 = 0.000000
FP26 = 0.000000
FP28 = 0.000000
FP30 = 0.000000

Console

Programmed By: Kunwar Arpit Singh 22185

Enter student name: Arpit
Enter total number of classes held: 40
Enter number of classes attended: 31
Arpit is allowed to sit in the exam.

[004000dc] 0000000c syscall ; 71: syscall
[004000e0] 08100041 j 0x00400104 [exit] ; 73: j exit
[004000e4] 34020004 ori \$2, \$0, 4 ; 76: li \$v0, 4
[004000e8] 3c011001 lui \$1, 4097 [student_name]; 77: la \$a0, student_name
[004000ec] 342400cc ori \$4, \$1, 204 [student_name]
[004000f0] 0000000c syscall ; 78: syscall
[004000f4] 34020004 ori \$2, \$0, 4 ; 80: li \$v0, 4
[004000f8] 3c011001 lui \$1, 4097 [not_allowed]; 81: la \$a0, not_allowed
[004000fc] 342400a7 ori \$4, \$1, 167 [not_allowed]
[00400100] 0000000c syscall ; 82: syscall
[00400104] 3402000a ori \$2, \$0, 10 ; 85: li \$v0, 10
[00400108] 0000000c syscall ; 86: syscall

FP Regs | **Int Regs [10]** | **Data** | **Text**

FP Regs

FIR = 38912
FCSR = 0

Single Precision

FG0 = 0.000000
FG1 = 0.000000
FG2 = 0.000000
FG3 = 0.000000
FG4 = 0.000000
FG5 = 0.000000
FG6 = 0.000000
FG7 = 0.000000
FG8 = 0.000000
FG9 = 0.000000
FG10 = 0.000000
FG11 = 0.000000
FG12 = 0.000000
FG13 = 0.000000
FG14 = 0.000000
FG15 = 0.000000
FG16 = 0.000000
FG17 = 0.000000
FG18 = 0.000000
FG19 = 0.000000
FG20 = 0.000000
FG21 = 0.000000
FG22 = 0.000000
FG23 = 0.000000
FG24 = 0.000000
FG25 = 0.000000
FG26 = 0.000000
FG27 = 0.000000
FG28 = 0.000000
FG29 = 0.000000
FG30 = 0.000000
FG31 = 0.000000

Double Precision

FP0 = 0.000000
FP2 = 0.000000
FP4 = 0.000000
FP6 = 0.000000
FP8 = 0.000000
FP10 = 0.000000
FP12 = 0.000000
FP14 = 0.000000
FP16 = 0.000000
FP18 = 0.000000
FP20 = 0.000000
FP22 = 0.000000
FP24 = 0.000000
FP26 = 0.000000
FP28 = 0.000000
FP30 = 0.000000

Console

Programmed By: Kunwar Arpit Singh 22185

Enter student name: Ajay
Enter total number of classes held: 40
Enter number of classes attended: 28
Ajay is not allowed to sit in the exam.

[004000dc] 0000000c syscall ; 71: syscall
[004000e0] 08100041 j 0x00400104 [exit] ; 73: j exit
[004000e4] 34020004 ori \$2, \$0, 4 ; 76: li \$v0, 4
[004000e8] 3c011001 lui \$1, 4097 [student_name]; 77: la \$a0, student_name
[004000ec] 342400cc ori \$4, \$1, 204 [student_name]
[004000f0] 0000000c syscall ; 78: syscall
[004000f4] 34020004 ori \$2, \$0, 4 ; 80: li \$v0, 4
[004000f8] 3c011001 lui \$1, 4097 [not_allowed]; 81: la \$a0, not_allowed
[004000fc] 342400a7 ori \$4, \$1, 167 [not_allowed]
[00400100] 0000000c syscall ; 82: syscall
[00400104] 3402000a ori \$2, \$0, 10 ; 85: li \$v0, 10
[00400108] 0000000c syscall ; 86: syscall

4. Problem 4

Problem statement

Write an MIPS assembly program that takes the marks of the student as an input (in the range of 1-100) and assign the grade. The grading policy are as follows:

Grade: A if marks ≥ 80
Grade: B if $80 < \text{marks} \leq 60$
Grade: C if $60 < \text{marks} \leq 40$
Grade: F otherwise

MIPS Assembly Source

```
1 # Assignment 7: Conditional Construct
2 # Programmed By : Kunwar Arpit Singh
3
4 .data
5     prompt_marks: .asciiz "Programmed By: Kunwar Arpit Singh 22185\n\nEnter
6         marks (0-100): "
7     grade_a: .asciiz "Grade: A\n"
8     grade_b: .asciiz "Grade: B\n"
9     grade_c: .asciiz "Grade: C\n"
10    grade_f: .asciiz "Grade: F\n"
11
12 .text
13 .globl main
14
15 main:
16     li $v0, 4
17     la $a0, prompt_marks
18     syscall
19
20     li $v0, 5
21     syscall
22     move $s0, $v0
23
24     li $t0, 40
25     blt $s0, $t0, grade_F
26
27     li $t0, 60
28     blt $s0, $t0, grade_C
29
30     li $t0, 80
31     blt $s0, $t0, grade_B
32
33 grade_A:
34     li $v0, 4
35     la $a0, grade_a
36     syscall
37     j exit
```

```
37
38 grade_B:
39     li $v0, 4
40     la $a0, grade_b
41     syscall
42     j exit
43
44 grade_C:
45     li $v0, 4
46     la $a0, grade_c
47     syscall
48     j exit
49
50 grade_F:
51     li $v0, 4
52     la $a0, grade_f
53     syscall
54
55 exit:
56     li $v0, 10
57     syscall
```

Output Screenshots

Integer Register values and Console

The image displays two screenshots of a debugger interface, showing the state of integer registers and the console output for a program.

Top Screenshot:

- Integer Registers (Int Regs [10]):**
 - PC = 4194480
 - EPC = 0
 - Cause = 0
 - BadVAddr = 0
 - Status = 805371664
 - HI = 0
 - LO = 0
 - R0 [r0] = 0
 - R1 [at] = 268500992
 - R2 [v0] = 10
 - R3 [v1] = 0
 - R4 [a0] = 268501055
 - R5 [a1] = 2147479220
 - R6 [a2] = 2147479240
 - R7 [a3] = 0
 - R8 [t0] = 80
 - R9 [t1] = 0
 - R10 [t2] = 0
 - R11 [t3] = 0
 - R12 [t4] = 0
 - R13 [t5] = 0
 - R14 [t6] = 0
 - R15 [t7] = 0
 - R16 [s0] = 81
 - R17 [s1] = 0
 - R18 [s2] = 0
 - R19 [s3] = 0
 - R20 [s4] = 0
 - R21 [s5] = 0
 - R22 [s6] = 0
 - R23 [s7] = 0
 - R24 [t8] = 0
 - R25 [t9] = 0
 - R26 [k0] = 0
 - R27 [k1] = 0
 - R28 [gp] = 268468224
 - R29 [sp] = 2147479216
 - R30 [s8] = 0
 - R31 [ra] = 4194328
- Console:**
 - Programmed By: Kunwar Arpit Singh 22185
 - Enter marks (0-100): 81
 - Grade: A
- Text (Disassembly):**
 - [00400090] 34240053 ori \$4, \$1, 83 [grade_c]
 - [00400094] 0000000c syscall ; 47: syscall
 - [00400098] 0810002b j 0x004000ac [exit] ; 48: j exit
 - [0040009c] 34020004 ori \$2, \$0, 4 ; 51: li \$v0, 4
 - [004000a0] 3c011001 lui \$1, 4097 [grade_f] ; 52: la \$a0, grade_f
 - [004000a4] 3424005d ori \$4, \$1, 93 [grade_f]
 - [004000a8] 0000000c syscall ; 53: syscall
 - [004000ac] 3402000a ori \$2, \$0, 10 ; 56: li \$v0, 10
 - [004000b0] 0000000c syscall ; 57: syscall

Bottom Screenshot:

- Integer Registers (Int Regs [10]):**
 - PC = 4194480
 - EPC = 0
 - Cause = 0
 - BadVAddr = 0
 - Status = 805371664
 - HI = 0
 - LO = 0
 - R0 [r0] = 0
 - R1 [at] = 268500992
 - R2 [v0] = 10
 - R3 [v1] = 0
 - R4 [a0] = 268501065
 - R5 [a1] = 2147479220
 - R6 [a2] = 2147479240
 - R7 [a3] = 0
 - R8 [t0] = 80
 - R9 [t1] = 0
 - R10 [t2] = 0
 - R11 [t3] = 0
 - R12 [t4] = 0
 - R13 [t5] = 0
 - R14 [t6] = 0
 - R15 [t7] = 0
 - R16 [s0] = 79
 - R17 [s1] = 0
 - R18 [s2] = 0
 - R19 [s3] = 0
 - R20 [s4] = 0
 - R21 [s5] = 0
 - R22 [s6] = 0
 - R23 [s7] = 0
 - R24 [t8] = 0
 - R25 [t9] = 0
 - R26 [k0] = 0
 - R27 [k1] = 0
 - R28 [gp] = 268468224
 - R29 [sp] = 2147479216
 - R30 [s8] = 0
 - R31 [ra] = 4194328
- Console:**
 - Programmed By: Kunwar Arpit Singh 22185
 - Enter marks (0-100): 79
 - Grade: B
- Text (Disassembly):**
 - [00400090] 34240053 ori \$4, \$1, 83 [grade_c]
 - [00400094] 0000000c syscall ; 47: syscall
 - [00400098] 0810002b j 0x004000ac [exit] ; 48: j exit
 - [0040009c] 34020004 ori \$2, \$0, 4 ; 51: li \$v0, 4
 - [004000a0] 3c011001 lui \$1, 4097 [grade_f] ; 52: la \$a0, grade_f
 - [004000a4] 3424005d ori \$4, \$1, 93 [grade_f]
 - [004000a8] 0000000c syscall ; 53: syscall
 - [004000ac] 3402000a ori \$2, \$0, 10 ; 56: li \$v0, 10
 - [004000b0] 0000000c syscall ; 57: syscall

FP Regs

Int Regs [10]

Data

Text

Int Regs [10]

x

Text

```

PC = 4194480
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664

HI = 0
LO = 0

R0 [r0] = 0
R1 [at] = 268500992
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 268501075
R5 [a1] = 2147479220
R6 [a2] = 2147479240
R7 [a3] = 0
R8 [t0] = 60
R9 [t1] = 0
R10 [t2] = 0
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 59
R17 [s1] = 0
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
R22 [s6] = 0
R23 [s7] = 0
R24 [t8] = 0
R25 [t9] = 0
R26 [k0] = 0
R27 [k1] = 0
R28 [gp] = 268468224
R29 [sp] = 2147479216
R30 [s8] = 0
R31 [ra] = 4194328

```

Console

Programmed By: Kunwar Arpit Singh 22185

Enter marks (0-100): 59

Grade: C

```

[00400090] 34240053 ori $4, $1, 83 [grade_c]
[00400094] 0000000c syscall ; 47: syscall
[00400098] 0810002b j 0x004000ac [exit] ; 48: j exit
[0040009c] 34020004 ori $2, $0, 4 ; 51: li $v0, 4
[004000a0] 3c011001 lui $1, 4097 [grade_f] ; 52: la $a0, grade_f
[004000a4] 3424005d ori $4, $1, 93 [grade_f]
[004000a8] 0000000c syscall ; 53: syscall
[004000ac] 3402000a ori $2, $0, 10 ; 56: li $v0, 10
[004000b0] 0000000c syscall ; 57: syscall

```

FP Regs

Int Regs [10]

Data

Text

Int Regs [10]

x

Text

```

PC = 4194480
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664

HI = 0
LO = 0

R0 [r0] = 0
R1 [at] = 268500992
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 268501085
R5 [a1] = 2147479220
R6 [a2] = 2147479240
R7 [a3] = 0
R8 [t0] = 40
R9 [t1] = 0
R10 [t2] = 0
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 39
R17 [s1] = 0
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
R22 [s6] = 0
R23 [s7] = 0
R24 [t8] = 0
R25 [t9] = 0
R26 [k0] = 0
R27 [k1] = 0
R28 [gp] = 268468224
R29 [sp] = 2147479216
R30 [s8] = 0
R31 [ra] = 4194328

```

Console

Programmed By: Kunwar Arpit Singh 22185

Enter marks (0-100): 39

Grade: F

```

[00400090] 34240053 ori $4, $1, 83 [grade_c]
[00400094] 0000000c syscall ; 47: syscall
[00400098] 0810002b j 0x004000ac [exit] ; 48: j exit
[0040009c] 34020004 ori $2, $0, 4 ; 51: li $v0, 4
[004000a0] 3c011001 lui $1, 4097 [grade_f] ; 52: la $a0, grade_f
[004000a4] 3424005d ori $4, $1, 93 [grade_f]
[004000a8] 0000000c syscall ; 53: syscall
[004000ac] 3402000a ori $2, $0, 10 ; 56: li $v0, 10
[004000b0] 0000000c syscall ; 57: syscall

```

Floating Point Register values and Console

The image displays two screenshots of a MIPS simulator interface, showing the state of floating point registers and the console output for different marks.

Top Screenshot:

- FP Regs:** Shows the state of floating point registers. The **Single Precision** registers (FG0 to FG31) are all set to 0.000000. The **Double Precision** registers (FP0 to FP30) are also all set to 0.000000.
- Console:** Displays the program output. It shows the marks entered (0-100): 81, resulting in a Grade: A.
- Assembly:** The assembly code is visible at the bottom, showing instructions for loading marks, calculating grades, and exiting.

Bottom Screenshot:

- FP Regs:** Shows the state of floating point registers. The **Single Precision** registers (FG0 to FG31) are all set to 0.000000. The **Double Precision** registers (FP0 to FP30) are also all set to 0.000000.
- Console:** Displays the program output. It shows the marks entered (0-100): 79, resulting in a Grade: B.
- Assembly:** The assembly code is visible at the bottom, showing instructions for loading marks, calculating grades, and exiting.

FP Regs

Int Regs [10]

Data

Text

FP Regs

FIR = 38912

FCSR = 0

Single Precision

FG0 = 0.000000
FG1 = 0.000000
FG2 = 0.000000
FG3 = 0.000000
FG4 = 0.000000
FG5 = 0.000000
FG6 = 0.000000
FG7 = 0.000000
FG8 = 0.000000
FG9 = 0.000000
FG10 = 0.000000
FG11 = 0.000000
FG12 = 0.000000
FG13 = 0.000000
FG14 = 0.000000
FG15 = 0.000000
FG16 = 0.000000
FG17 = 0.000000
FG18 = 0.000000
FG19 = 0.000000
FG20 = 0.000000
FG21 = 0.000000
FG22 = 0.000000
FG23 = 0.000000
FG24 = 0.000000
FG25 = 0.000000
FG26 = 0.000000
FG27 = 0.000000
FG28 = 0.000000
FG29 = 0.000000
FG30 = 0.000000
FG31 = 0.000000

Double Precision

FP0 = 0.000000
FP2 = 0.000000
FP4 = 0.000000
FP6 = 0.000000
FP8 = 0.000000
FP10 = 0.000000
FP12 = 0.000000
FP14 = 0.000000
FP16 = 0.000000
FP18 = 0.000000
FP20 = 0.000000
FP22 = 0.000000
FP24 = 0.000000
FP26 = 0.000000
FP28 = 0.000000
FP30 = 0.000000

Console

Programmed By: Kunwar Arpit Singh 22185

Enter marks (0-100): 59

Grade: C

```

[00400084] 0810002b j 0x004000ac [exit] ; 42: j exit
[00400088] 34020004 ori $2, $0, 4 ; 45: li $v0, 4
[0040008c] 3c011001 lui $1, 4097 [grade_c] ; 46: la $a0, grade_c
[00400090] 34240053 ori $4, $1, 83 [grade_c]
[00400094] 0000000c syscall ; 47: syscall
[00400098] 0810002b j 0x004000ac [exit] ; 48: j exit
[0040009c] 34020004 ori $2, $0, 4 ; 51: li $v0, 4
[004000a0] 3c011001 lui $1, 4097 [grade_f] ; 52: la $a0, grade_f
[004000a4] 3424005d ori $4, $1, 93 [grade_f]
[004000a8] 0000000c syscall ; 53: syscall

```

FP Regs

Int Regs [10]

Data

Text

FP Regs

FIR = 38912

FCSR = 0

Single Precision

FG0 = 0.000000
FG1 = 0.000000
FG2 = 0.000000
FG3 = 0.000000
FG4 = 0.000000
FG5 = 0.000000
FG6 = 0.000000
FG7 = 0.000000
FG8 = 0.000000
FG9 = 0.000000
FG10 = 0.000000
FG11 = 0.000000
FG12 = 0.000000
FG13 = 0.000000
FG14 = 0.000000
FG15 = 0.000000
FG16 = 0.000000
FG17 = 0.000000
FG18 = 0.000000
FG19 = 0.000000
FG20 = 0.000000
FG21 = 0.000000
FG22 = 0.000000
FG23 = 0.000000
FG24 = 0.000000
FG25 = 0.000000
FG26 = 0.000000
FG27 = 0.000000
FG28 = 0.000000
FG29 = 0.000000
FG30 = 0.000000
FG31 = 0.000000

Double Precision

FP0 = 0.000000
FP2 = 0.000000
FP4 = 0.000000
FP6 = 0.000000
FP8 = 0.000000
FP10 = 0.000000
FP12 = 0.000000
FP14 = 0.000000
FP16 = 0.000000
FP18 = 0.000000
FP20 = 0.000000
FP22 = 0.000000
FP24 = 0.000000
FP26 = 0.000000
FP28 = 0.000000
FP30 = 0.000000

Console

Programmed By: Kunwar Arpit Singh 22185

Enter marks (0-100): 39

Grade: F

```

[00400084] 0810002b j 0x004000ac [exit] ; 42: j exit
[00400088] 34020004 ori $2, $0, 4 ; 45: li $v0, 4
[0040008c] 3c011001 lui $1, 4097 [grade_c] ; 46: la $a0, grade_c
[00400090] 34240053 ori $4, $1, 83 [grade_c]
[00400094] 0000000c syscall ; 47: syscall
[00400098] 0810002b j 0x004000ac [exit] ; 48: j exit
[0040009c] 34020004 ori $2, $0, 4 ; 51: li $v0, 4
[004000a0] 3c011001 lui $1, 4097 [grade_f] ; 52: la $a0, grade_f
[004000a4] 3424005d ori $4, $1, 93 [grade_f]
[004000a8] 0000000c syscall ; 53: syscall

```